

Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

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Abstract

This work investigates the use of Abaqus-native error-guided mesh refinement for phase-field fracture simulations implemented with user elements. The point of departure is the staggered phase-field implementation of Molnár and Gravouil and the mesh-refinement procedure proposed by Pandey and Kumar. The first objective is to reproduce reference fracture simulations without mesh refinement; the second is to implement the refinement procedure in Python while retaining the layered UEL/UMAT representation required by the available phase-field code; the third is to assess the additional requirements introduced by nonmatching state transfer and changing mesh topology.

A fixed-mesh Mode-I single-edge-crack calculation was reproduced before introducing adaptive refinement. The reconstructed response reached a maximum reaction force of 0.757778 kN at $u = 0.005857$ mm. The project reference curve used for comparison gives approximately 0.758 kN at $u \approx 0.005860$ mm, corresponding to differences of about -0.029% in peak load and -0.051% in peak displacement. This fixed-mesh calculation therefore provides the current quantitative baseline for the refinement study.

A Python-controlled two-job workflow was then implemented following Pandey and Kumar. A coarse pre-analysis supplies the Abaqus recovered-stress discretization indicator `MISESeri`; an Abaqus `RemeshingRule` and `adaptiveRemesh` operation generate a refined physical mesh; and the resulting connectivity is converted automatically into the coincident phase-field, mechanical and visualization layers used by the user-element formulation. A headless Abaqus qualification demonstrated genuine native remeshing and exposed an implementation detail that had initially overwritten the adaptively generated mesh by calling `generateMesh()` a second time.

The authoritative Task-5 production calculation (`1399632.mm``master02`) on the corrected sharp-slit 71,320-element adaptive mesh completed technically to the requested displacement endpoint $u = 0.010$ mm in 35 h 08 min of walltime. The calculation resolved the post-peak crack propagation branch to a final load drop of 94.4% with a pure Mode-I horizontal crack path. However, the calculation failed the quantitative benchmark reproduction gates: the extracted peak load is 0.478203 kN at $u = 0.004150$ mm (-36.89% in peak force and -29.14% in peak displacement vs baseline), initial elastic stiffness is reduced by -11.3% , and the native 1% error target generated 71,320 elements versus $\approx 13,941$ reported in the reference paper. A dedicated discrepancy audit confirmed that the UEL formulation, seam boundary conditions, and property vectors are mathematically sound, while the quantitative divergence reflects a combined structural compliance shift and earlier tip damage localization on the over-refined discretization. Task 5 remains open pending quantitative resolution.

In parallel, nonmatching transfer of displacement, phase field and history variables was investigated. Runtime state ingestion was implemented successfully, but topology-changing restart calculations showed reaction-force changes of approximately 26.3% and 99.6% during the relaxed restart stage for two tested crack-tip states. The current UEL also does not populate the Abaqus `ENERGY` array, so a directly comparable internal-energy continuity check is unavailable.

Consequently, technically successful state installation is distinguished from mechanically consistent evolving-remesh continuation. The present shared-state implementation is qualified only for serial production execution because unsynchronised accesses to mutable `COMMON` data prevent a defensible threaded or MPI production claim.

Chapter 1

Problem Statement, Theoretical Basis, and Reference Methods

1.1 Problem statement

Phase-field fracture regularizes a sharp crack over a finite length scale and thereby avoids explicit tracking of a discontinuity. The numerical advantage is accompanied by a spatial-resolution requirement: the diffuse crack zone must be resolved by the finite-element mesh. A uniformly fine mesh is therefore expensive when only a small part of the structure develops damage. The thesis addresses whether Abaqus built-in mesh-refinement capabilities can be combined with a phase-field UEL/UMAT formulation so that fine resolution is concentrated in mechanically relevant regions without changing the underlying fracture model.

The work is organised around four interfaces that are kept explicit throughout the report:

1. **Phase-field constitutive model:** equations and state variables implemented by the supplied user subroutines.
2. **Abaqus error estimation and mesh generation:** the standard-element/facsimile representation used to expose stresses and MISESERR to Abaqus.
3. **UEL reconstruction:** conversion of the refined physical mesh into the coincident user-element and visualization layers required by the phase-field implementation.
4. **State transfer and restart:** transfer of displacement, phase field and history variables when the discretization changes during a continued analysis.

This separation is important because a correct result at one interface does not imply correctness of the complete workflow.

1.2 Phase-field fracture formulation

The implementation follows the standard AT2-type regularization used in the reference phase-field literature (Msekh et al., 2015; Molnár and Gravouil, 2017; Pandey and Kumar, 2025). For displacement field $\mathbf{u}$ and phase field $d \in [0, 1]$, with $d = 0$ intact and $d = 1$ fully broken,

the total potential can be written in the generic form

$$\Pi(\mathbf{u}, d) = \int_{\Omega} \left[g(d) \psi_0^+(\boldsymbol{\varepsilon}) + \psi_0^-(\boldsymbol{\varepsilon}) \right] d\Omega + \int_{\Omega} G_c \gamma(d, \nabla d) d\Omega - W_{\text{ext}}, \quad (1.1)$$

where G_c is the fracture energy, $g(d)$ is the degradation function, and γ is the regularized crack-density functional. For the AT2 choice used in the reference work,

$$\gamma(d, \nabla d) = \frac{1}{2} \left(\frac{d^2}{l_0} + l_0 |\nabla d|^2 \right), \quad (1.2)$$

with regularization length l_0 . A small residual stiffness is retained through a modified quadratic degradation function, typically $g(d) = (1 - d)^2 + k$ with $k \ll 1$.

Crack irreversibility is enforced through a history field

$$\mathcal{H}(\mathbf{x}, t) = \max_{\tau \leq t} \psi_0^+(\boldsymbol{\varepsilon}(\mathbf{x}, \tau)), \quad (1.3)$$

so that the crack-driving quantity is non-decreasing. In this thesis, monotonicity of the stored history variable and monotonicity of projected visualization variables are evaluated separately. A decrease in a projected state variable is not automatically interpreted as constitutive healing.

1.3 Abaqus UEL/UMAT architecture

The supplied implementation follows the staggered Abaqus strategy described by Molnár and Gravouil (Molnár and Gravouil, 2017). The physical discretization is represented by coincident logical layers. User elements solve the phase-field and mechanical problems, while companion standard elements driven through a UMAT expose stresses and solution-dependent variables to the Abaqus output database. The mapping between these layers is part of the numerical method and is checked explicitly after every regenerated mesh.

The main data path relevant to refinement is

physical deformation $\rightarrow$ mechanical UEL $\rightarrow$ shared state $\rightarrow$ UMAT/facsimile layer $\rightarrow$ Abaqus
stress recovery $\rightarrow$ MISESeri.

Thus MISESeri is a recovered-stress discretization indicator. It is not a phase-field error estimator and is used here only to identify where the coarse mechanical solution has strong unresolved gradients.

The coincident-layer architecture is verified in the input-deck connectivity and element-type audits. A CAE layer excerpt is not inserted here because overlapping layers are not visually separable in the available viewport export; the report does not substitute an illustrative rendering for that model evidence.

1.4 Reference mesh-refinement method

Pandey and Kumar proposed a Python-controlled two-job procedure for phase-field calculations in Abaqus (Pandey and Kumar, 2025). Their script defines a `RemeshingRule` on the facsimile set `All_elem`, requests MISESeri, and subsequently calls `adaptiveRemesh(odb=...)` using the pre-analysis output database. For the Mode-I edge-crack example, they report a global

coarse size of 0.02 mm, local refined size 0.001 mm, $l_0 = 0.0075$ mm, $G_c = 2.7 \times 10^{-3}$ kN/mm, and 13,941 elements for the proposed mesh. Their Listing 1 shows `errorTarget=1.0`, `refinementFactor=10`, and no coarsening.

The method is therefore described in this report as *Abaqus-native error-guided pre-refinement*. The final fracture calculation uses a fixed refined mesh. This is distinct from automated external-driver remeshing during crack propagation, and from true in-analysis remeshing.

1.5 Objectives of the present work

The work reported here addresses the following questions:

1. Can the supplied phase-field implementation reproduce a reference fixed-mesh Mode-I response?
2. Can the Pandey–Kumar Abaqus-native refinement procedure be reproduced in a headless Python workflow and converted reliably to the layered UEL/UMAT formulation?
3. Does the refined model reproduce the published force-displacement response and mesh density?
4. What information must be transferred when the mesh changes during an ongoing phase-field calculation, and does the restarted model preserve mechanical continuity?
5. Which aspects of the current UEL/UEXTERNALDB architecture limit parallel execution and future ABAQUSER integration?

Chapter 2

Verification of the Supplied Phase-Field Implementation

2.1 Purpose of the baseline

Before introducing mesh refinement, the phase-field implementation must reproduce a problem for which the expected response is known. This chapter therefore separates source-code verification from mesh-adaptation questions. The Molnár–Gravouil implementation provides the staggered UEL/UMAT basis (Molnár and Gravouil, 2017), while the finite-size edge-crack Mode-I benchmark of Pandey and Kumar provides the principal quantitative comparison used in the current refinement study (Pandey and Kumar, 2025).

2.2 Molnár one-element verification

The unchanged one-element example supplied with the Molnár–Gravouil implementation was compiled and executed before any fracture benchmark or remeshing modification. The archived ODB contains 1,001 frames in the step `Static` and the expected displacement, reaction-force and state-variable outputs, including `SDV15`. Figure 2.1 is a direct Abaqus CAE export from that ODB. Its purpose is deliberately narrow: it establishes that the original user-element source executes and evolves its constitutive state before the code is exercised on a crack problem.

2.3 Molnár–Gravouil single-notch reproduction

The next verification level was the author-supplied single-edge-notched specimen. Figure 2.2 shows the actual mesh reconstructed from the unchanged input deck, including the refined crack corridor, initial notch and applied boundary conditions. This calculation provides the first direct crack-evolution test of the supplied staggered UEL/UMAT implementation.

The phase-field history in Figure 2.3 uses four actual converged ODB states. The displayed quantity is `SDV15`, interpreted as $d = 0$ intact and $d = 1$ fully damaged. The common color scale and viewport make the progression from diffuse notch-tip damage to a connected crack directly visible.

Execution of the original supplied model did not imply exact reproduction of the published

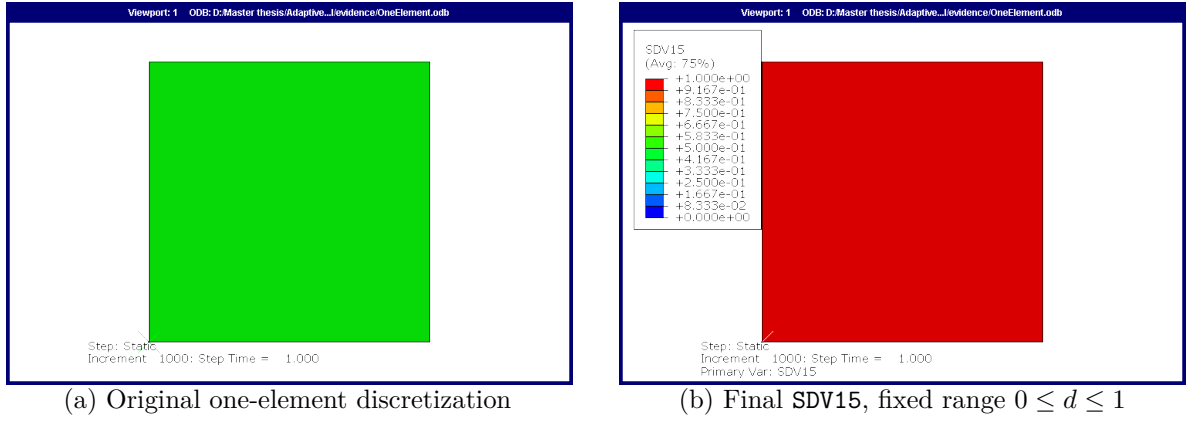


Figure 2.1: Direct Abaqus CAE evidence from the unchanged Molnár one-element ODB. The image establishes source execution and state evolution; it is not presented as a fracture benchmark.

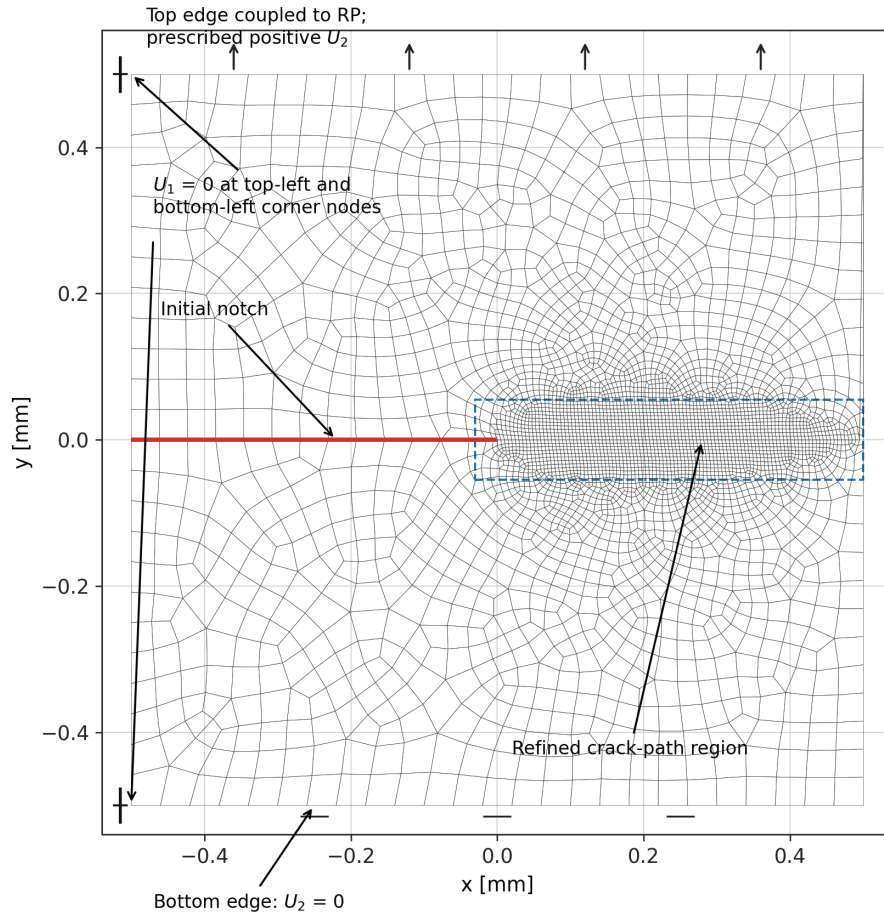


Figure 2.2: Actual Molnár author-supplied single-notch mesh and boundary conditions. The mesh was parsed from the unchanged input deck; no illustrative geometry was substituted.

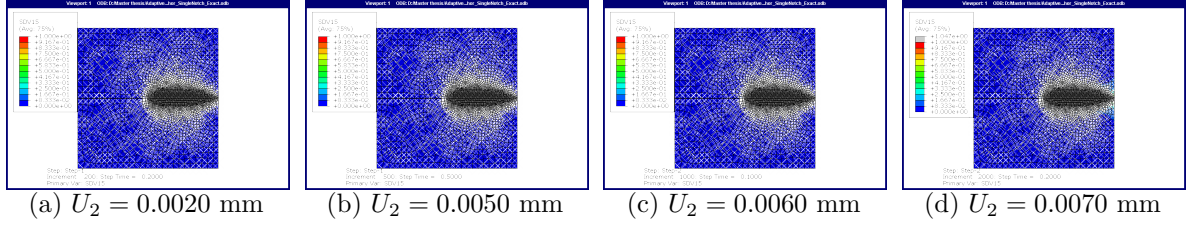


Figure 2.3: Direct Abaqus CAE exports of Molnár single-notch SDV15 at four converged displacement states. Every panel uses the fixed range $0 \leq d \leq 1$ and the same viewport.

curve. The simulation peak was 0.727608 kN at $U_2 = 0.006100$ mm, while the digitized Fig. 7 reference peak was 0.690591 kN near 0.005584 mm. The corresponding peak-force and peak-displacement differences were approximately 5.36% and 9.25%, respectively. Figure 2.4 therefore reports the mismatch explicitly.

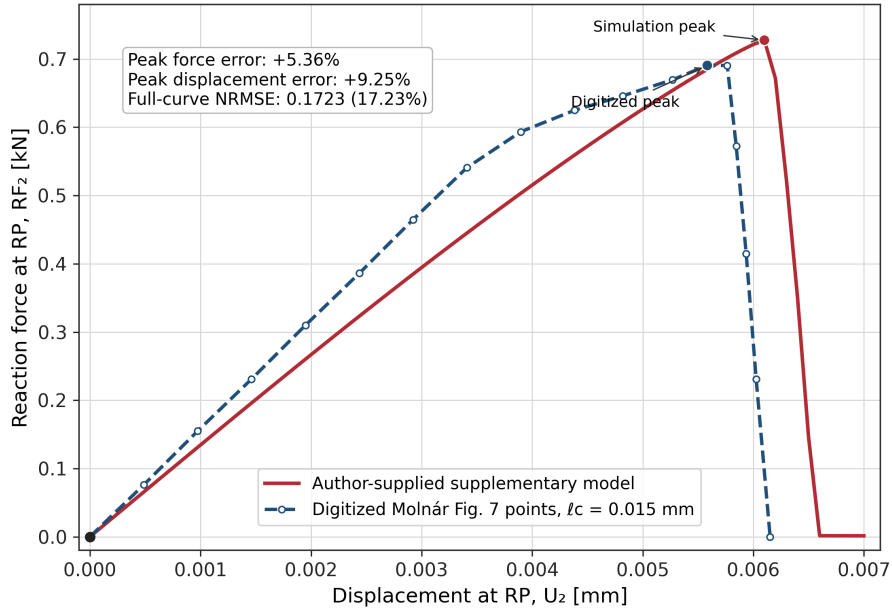


Figure 2.4: Reaction-force-displacement comparison between the unchanged author-supplied Molnár Abaqus model and the digitized $l_c = 0.015$ mm curve from Molnár and Gravouil.

2.4 Improved Molnár Mode-I reference

The initial discrepancy motivated a controlled mesh-resolution study rather than being hidden or treated as a remeshing effect. Figure 2.6 summarizes the later H0/H1/H2 progression and shows how response convergence and computational cost were evaluated together. This study established the stronger fixed-mesh basis used by the subsequent refinement work.

2.5 Fixed-mesh Mode-I benchmark for the Pandey–Kumar comparison

The benchmark is a 1×1 mm square plate with a 0.5 mm edge crack. The material parameters are $E = 210$ GPa and $\nu = 0.3$. The fracture parameters are $l_0 = 0.0075$ mm and $G_c =$

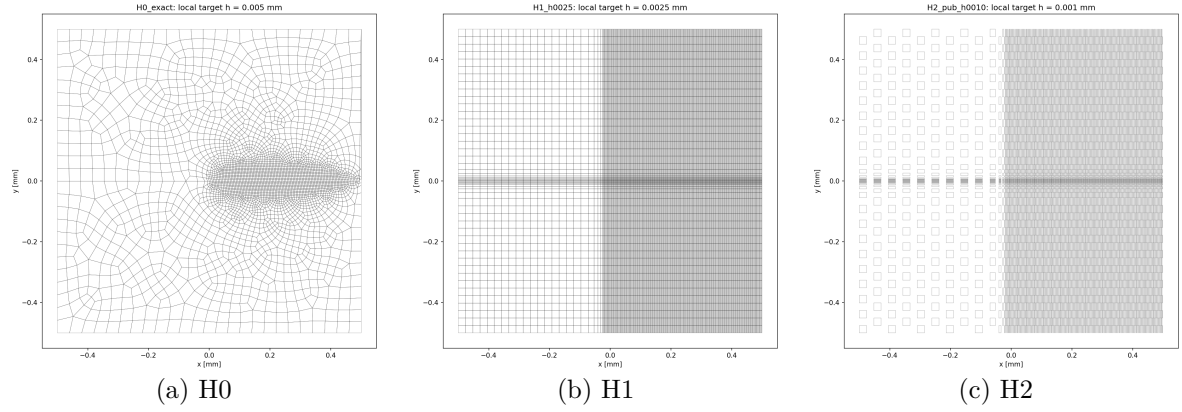


Figure 2.5: Actual H0/H1/H2 Molnár meshes generated by the project preprocessing chain. The panels expose the discretizations behind the convergence data in Figure 2.6.

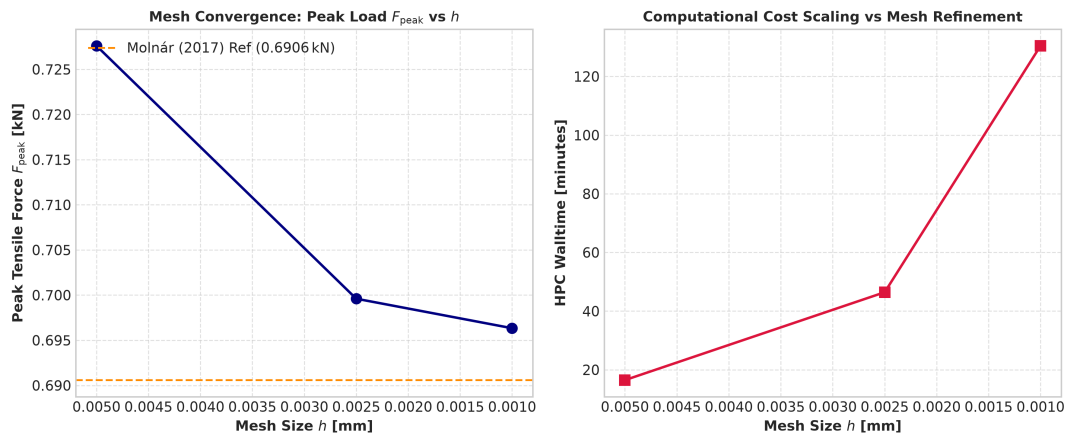


Figure 2.6: Molnár Mode-I mesh-resolution study. Response agreement is assessed together with mesh density and computational cost rather than from a single peak value.

2.7×10^{-3} kN/mm. Loading is displacement controlled. In the publication, the standard phase-field calculation uses a global mesh of 0.02 mm and a locally refined crack corridor with $h = 0.003$ mm; the displacement is applied in two steps up to 0.005 mm and then 0.010 mm (Pandey and Kumar, 2025).

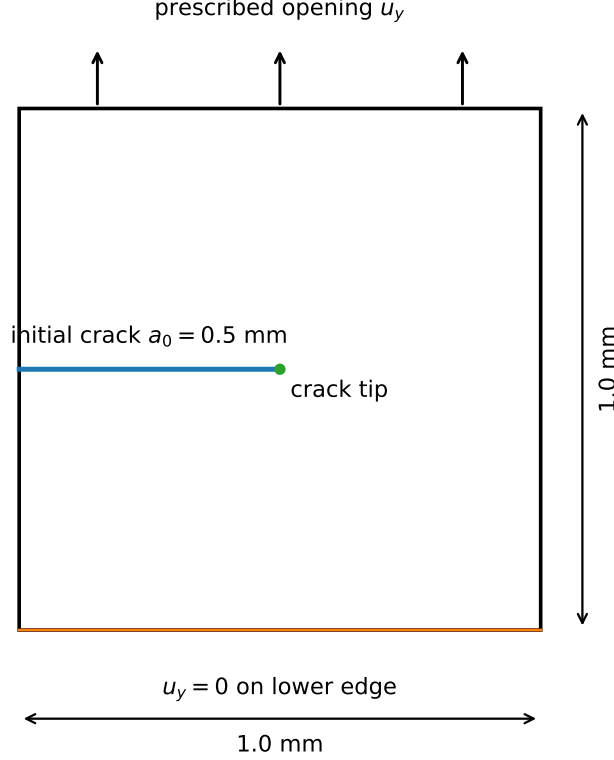


Figure 2.7: Schematic of the Mode-I single-edge-crack benchmark used for the fixed-mesh reproduction and the subsequent mesh-refinement study. The crack is represented as a zero-gap discontinuity; this geometric detail is important for the elastic compliance of the specimen.

2.6 Fixed-mesh quantitative reproduction

The project fixed-mesh reference calculation reached

$$F_{\max} = 0.757778 \text{ kN}, \quad u(F_{\max}) = 0.005857 \text{ mm}. \quad (2.1)$$

The digitized/project reference values used for the comparison are approximately 0.758 kN and 0.005860 mm. Table 2.1 summarizes the scalar comparison.

Table 2.1: Fixed-mesh Mode-I reproduction compared with the project reference values extracted from the published benchmark.

Quantity	Reference	Reproduction	Relative difference
Peak force $F_{\max}$	0.758000 kN	0.757778 kN	−0.029%
Displacement at peak	0.005860 mm	0.005857 mm	−0.051%

The complete force–displacement curve remains the preferred comparison quantity. The peak values are reported here because they provide a compact check that the reconstructed

benchmark has the correct force scale and localization displacement before changing the mesh-generation procedure.

The accepted standard ODB also contains the matched-state fields required for a final standard-versus-adaptive comparison. That combined figure is deferred until the adaptive calculation provides verified terminal states at matching displacements.

2.7 Geometry sanity check for later adaptive models

The baseline was also used as a model-level diagnostic. A later refinement attempt accidentally represented the pre-crack as a finite-width rectangular slot. Its initial stiffness was 75.47 kN/mm, whereas the accepted fixed-mesh baseline was about 138.15 kN/mm. Replacing the slot by a zero-gap seam restored the pre-analysis stiffness to 141.82 kN/mm, a difference of about +2.65% from the baseline. This comparison identified a geometry error before interpreting the discrepancy as a phase-field or remeshing effect.

2.8 Baseline conclusion

The fixed-mesh Mode-I response is sufficiently well reproduced to serve as the current reference for the refined calculation. The later comparison therefore considers the complete force–displacement curve, peak force, displacement at peak, crack path and computational cost. Agreement of a single scalar quantity is not by itself considered sufficient validation.

Chapter 3

Abaqus-Native Error-Guided Mesh Refinement

3.1 Role of MISESeri

Abaqus provides recovered-stress error indicators for standard continuum elements. In the present workflow, the companion UMAT elements reproduce the mechanical fields of the user-element discretization so that Abaqus can evaluate **MISESeri**. The indicator measures stress-discretization error associated with the recovered von Mises stress field; it is not a measure of phase-field discretization error. Its role is therefore predictive: it identifies mechanically under-resolved regions in the preliminary solution that are plausible candidates for subsequent fracture localization.

3.2 Reference-faithful two-job workflow

The implemented Python sequence follows the structure reported by Pandey and Kumar (Pandey and Kumar, 2025):

1. create the coarse physical model and the coincident output/facsimile layer;
2. run the preliminary analysis and request **MISESeri**, **MISESAVG**, stress, volume, reaction force and displacement;
3. define an Abaqus **RemeshingRule** on the assembly set corresponding to **All_elem**;
4. open the preliminary ODB and execute `model.adaptiveRemesh(odb=odb)`;
5. export the newly refined physical mesh;
6. reconstruct the phase-field UEL layer, mechanical UEL layer and visualization/UMAT layer on the new connectivity;
7. audit node coordinates, element connectivity, sets, boundary conditions, element types and property assignments before the fracture solve.

A headless qualification revealed an important Abaqus scripting detail. The operation **adaptiveRemesh** already modifies the model mesh. Calling **generateMesh()** afterwards reapplied the original part seeds and overwrote the refined mesh. Removing that second meshing

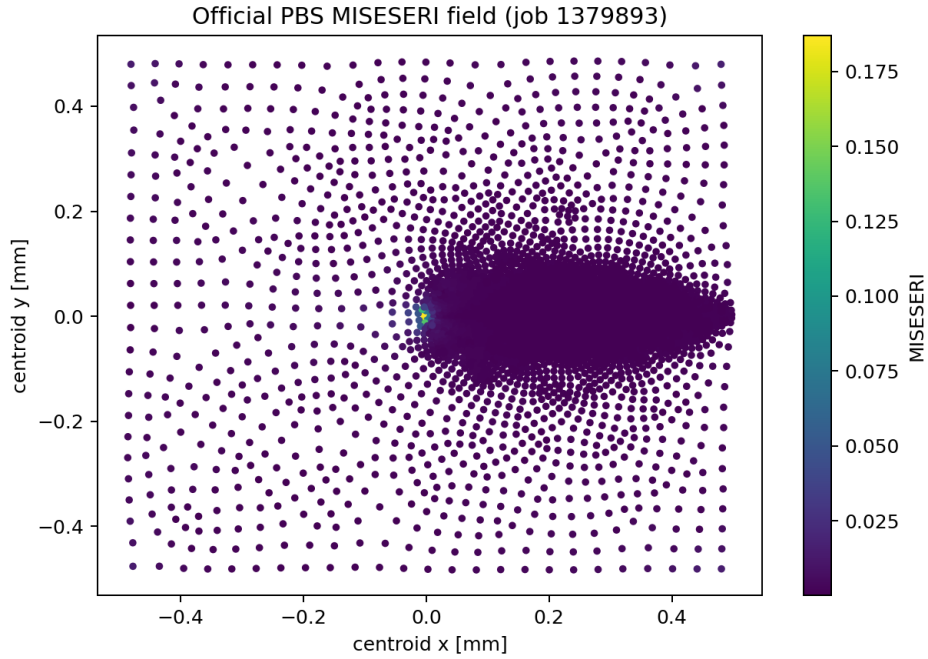


Figure 3.1: Spatial distribution of the official PBS-extracted MisesERI field from coarse pre-analysis job 1379893.mm_{master02}. The plot is generated from the 3,930-row extraction CSV with SHA-256 49b0c5f7a784f361...; the ODB, CSV and repaired validation are recorded together in the Stage-F4 replacement evidence. This is a derived plot of Abaqus element-field output, not a reconstructed phase-field contour.

Abaqus-native error-guided pre-refinement workflow

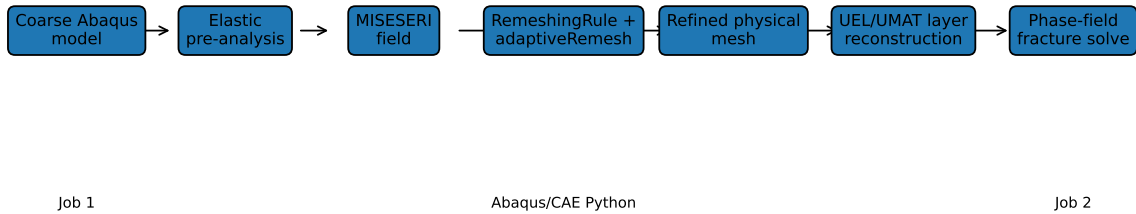


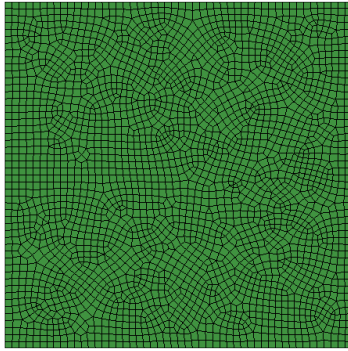
Figure 3.2: Computational sequence used to reproduce the Pandey–Kumar error-guided pre-refinement method. The fracture calculation begins from a virgin state on the final refined mesh; state transfer is not part of this two-job pre-refinement workflow.

call changed the test problem from an unchanged 553-element mesh to a genuinely refined 2,222-element mesh. This check was added as a regression condition: an adaptive-remesh operation that leaves topology unchanged when nontrivial MISESeri requires refinement is treated as a failed qualification.

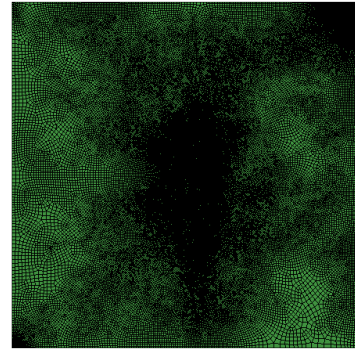
3.3 Mode-I reference settings

For the principal Mode-I case, Pandey and Kumar report a global element size of 0.02 mm and a local element size of 0.001 mm (Pandey and Kumar, 2025). The remeshing rule uses `errorTarget=1.0` and `refinementFactor=10`, with coarsening disabled. Their refined Mode-I mesh contains 13,941 triangular and quadrilateral elements. The present reconstruction uses these nominal settings together with the corrected sharp seam.

Figure 3.3 provides direct CAE mesh evidence for the two ends of the implemented workflow. The sensitivity cases remain represented by their audited counts until their distinct mesh artifacts can be exported with the same viewport; no reconstructed pattern is substituted.



(a) Coarse pre-analysis mesh



(b) Nominal 1% refined physical mesh

Figure 3.3: Direct Abaqus CAE mesh exports from the coarse pre-analysis input and the accepted nominal refined physical input. The dense central corridor in (b) is the actual native-remeshing result, not an illustrative reconstruction.

3.4 Observed mesh-density discrepancy

The reconstructed 1% case produced 71,320 physical elements, substantially more than the published 13,941. Because a model may solve correctly while still failing to reproduce the reference refinement pattern, a separate preprocessing sensitivity study was performed without changing the fracture parameters. For the current reconstruction, 2% and 5% error targets generated 15,396 and 4,194 elements, respectively.

The sensitivity study demonstrates that the remeshing outcome is highly sensitive to the interpretation of the error target and the resulting Abaqus sizing field. It does *not* establish that the publication used 2% in the Mode-I result. In fact, the paper explicitly lists `errorTarget=1.0` in its remeshing-rule function and separately reports a later sensitivity study with other error-target values. The origin of the remaining element-count discrepancy must therefore be stated as unresolved. Plausible contributors to examine include the precise pre-analysis loading history, the spatial region attached to the remeshing rule, Abaqus release-dependent sizing behaviour, and details of the coarse-mesh topology that are not completely recoverable from the publication.

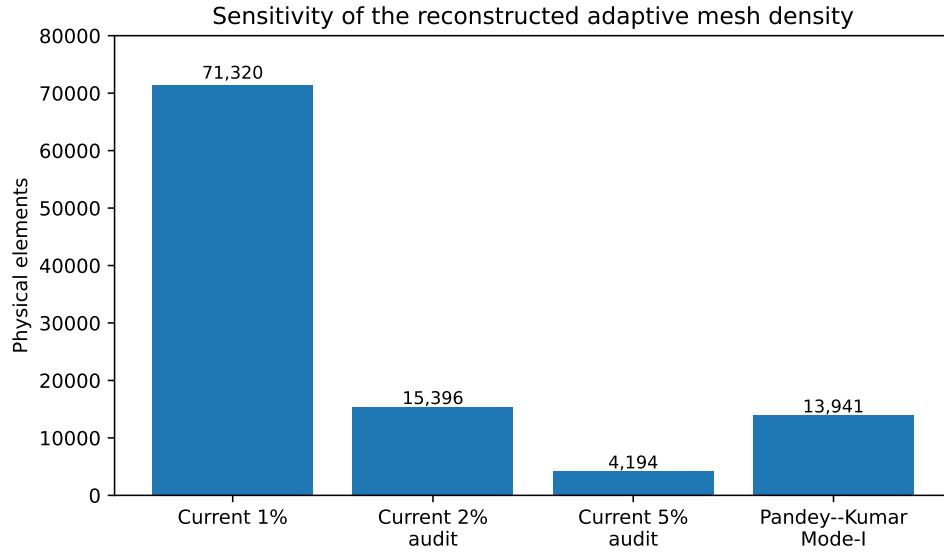


Figure 3.4: Physical-element counts obtained from the present Abaqus-native refinement reconstruction. The 2% and 5% cases are sensitivity calculations, not replacements for the nominal 1% production setting. The publication reports 13,941 elements for its Mode-I proposed mesh.

3.5 Interpretation

Task 4 is considered implemented at the level of the execution chain: Abaqus-native refinement runs headlessly, produces a genuinely changed mesh, and the result can be converted into the layered UEL/UMAT model with deterministic connectivity checks. Task 5 remains a quantitative reproduction problem because the refined mesh density and the final fracture response must both be compared with the publication. The current 1% fracture calculation is discussed in Chapter ?? and is still running at the time of report preparation.

Chapter 4

Nonmatching State Transfer and Restart

4.1 Why state transfer is required

Pre-refinement does not require state transfer because the final fracture calculation starts from the virgin state. Evolving remeshing is different: when the mesh changes after damage has developed, the receiving discretization must recover the mechanical displacement, phase field and irreversible history state of the donor analysis. A technically valid interpolation is not sufficient; the restarted model must also remain mechanically and thermodynamically consistent.

4.2 Transferred quantities and mapping spaces

The implemented transfer path distinguishes nodal and integration-point quantities. Displacement and phase field are transferred in nodal space using donor-element search and isoparametric interpolation. The crack-driving history field is stored at integration points and is therefore transferred with a separate Gauss-point mapping procedure. Crack flanks are treated as separate topological regions to avoid interpolation across an existing slit.

The receiving state is checked for phase bounds and history positivity. For irreversible fracture, preservation of the donor history is more important than reproducing a visually smooth field because a reduced history value can permit unphysical damage recovery.

4.3 Runtime state ingestion

An early transfer attempt generated a mapping artifact correctly but did not alter the running solution because the active UEL initialization path did not consume the prepared state. This distinction led to a redesign in which the transfer data are loaded explicitly at runtime through `UEXTERNALDB` and shared with the user-element routines. Subsequent tests confirmed that the transferred displacement, phase and history data were actually installed into the receiver analysis.

Figure 4.1 shows the actual checkpoint phase and history fields used by the transfer workflow. It complements the numerical transfer tables by making the donor localization state visible.

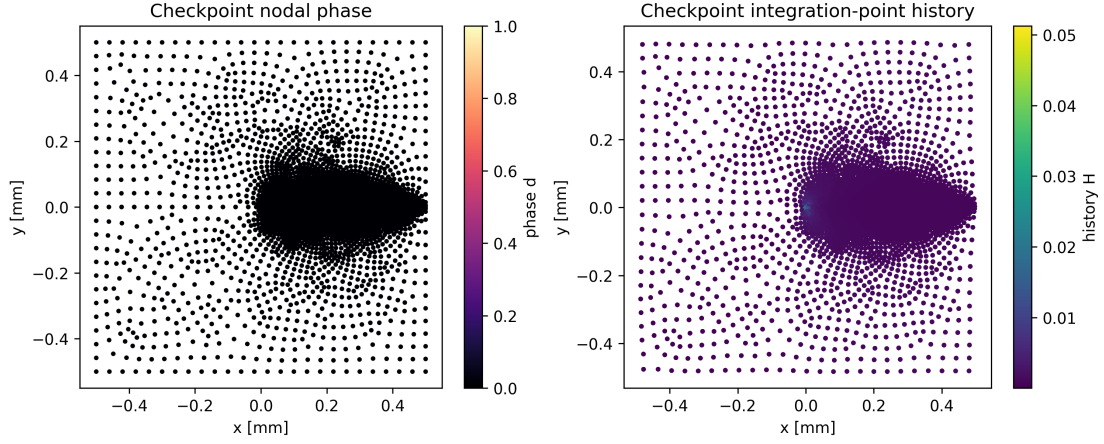


Figure 4.1: Checkpoint phase-field and history-field evidence used by the nonmatching state-transfer workflow. The panels originate from extracted simulation fields at the same donor checkpoint.

4.4 Restart sequence

To separate state installation from subsequent equilibration, the receiver calculation is divided conceptually into four stages:

1. install the transferred state and prescribed donor kinematics;
2. establish mechanical equilibrium while the phase field is held fixed;
3. release the phase-field constraints and allow the coupled state to relax;
4. continue the external loading.

This sequence improved numerical robustness, but it must not be interpreted as a guarantee of physical continuity. The decisive comparison is the state after the phase field has been released and relaxed.

4.5 Topology-changing restart results

Two topology-changing Mode-II restart calculations reached the continuation stage but failed the predeclared reaction-force continuity check after phase release. A transfer based on an earlier donor state exhibited a force reduction of approximately 26.31%. A later donor state with a more strongly degraded crack-tip facet exhibited a reduction of approximately 99.56%.

The later case is particularly informative: the geometric facet selected for separation carried only a small local traction before the split, yet releasing the phase field caused a much larger structural response. This shows that a local crack-tip criterion alone is insufficient to establish restart equivalence; the receiving phase-field problem can reorganize over a larger process zone.

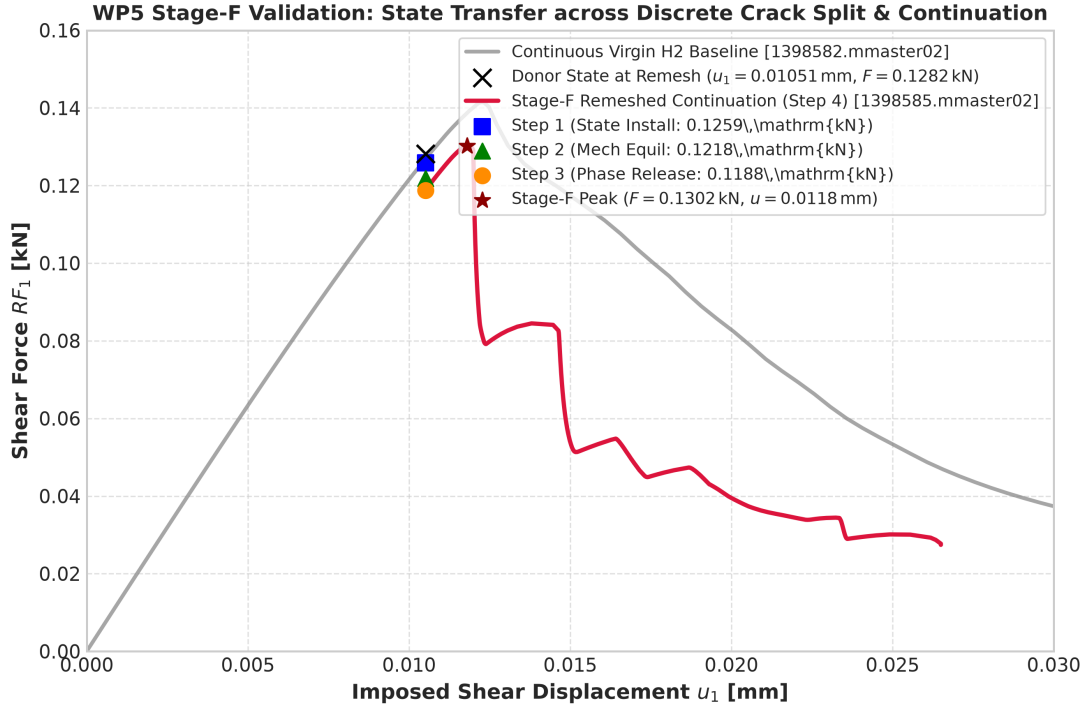


Figure 4.2: Topology-changing transfer and continuation evidence across state installation, equilibration, phase release and continuation. The figure visualizes why successful ingestion and solver continuation do not by themselves establish restart equivalence.

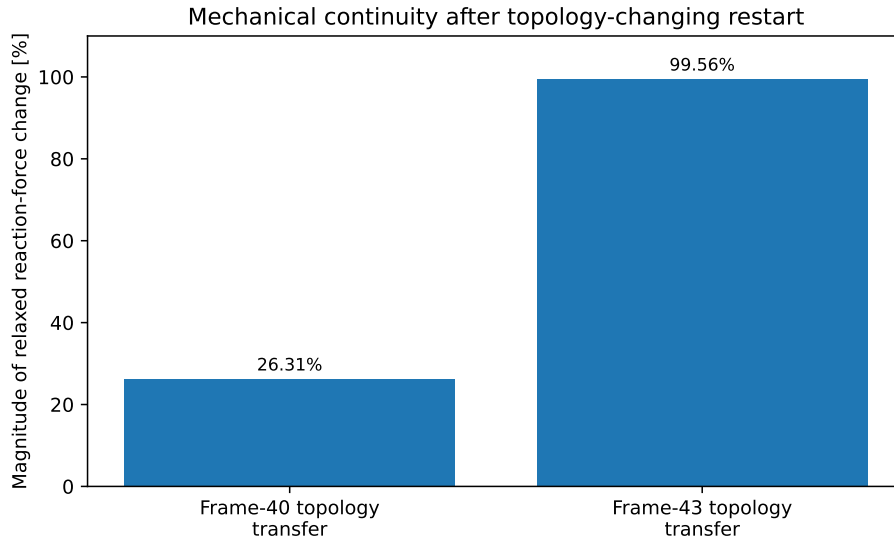


Figure 4.3: Magnitude of the reaction-force change between the mechanically equilibrated receiver state and the relaxed phase-release state for two tested topology-changing restarts. The results demonstrate that successful solver continuation is not equivalent to mechanically continuous state transfer.

4.6 Energy-accounting limitation

The current UEL includes the Abaqus `ENERGY(8)` argument but does not populate a qualified internal-energy contribution. Consequently, global Abaqus variables such as `ALLSE` and `ALLIE` do not provide a complete UEL-aware fracture-energy balance for these restart comparisons. External work or $\frac{1}{2}Fu$ estimates are therefore not substituted for the missing internal-energy continuity check.

4.7 Current conclusion

The work demonstrates nonmatching data mapping and actual runtime ingestion of the transferred variables. It does not yet demonstrate a topology-changing restart that preserves the relaxed mechanical state. This distinction is retained in the remainder of the thesis: state-transfer infrastructure is available, while mechanically consistent evolving remeshing remains an open problem.

Chapter 5

Multi-Resolution and Mode-II Validation Studies

5.1 Purpose of the secondary benchmarks

The Mode-II and multi-resolution studies were introduced to stress-test the transfer and refinement infrastructure under a curved or shear-dominated crack path. They are secondary to the main Pandey–Kumar Mode-I reproduction and are therefore presented here as validation studies rather than as substitutes for the mandatory mesh-refinement benchmark.

5.2 Uniform Mode-II mesh study

Three uniform discretizations were previously analysed with 3,930, 12,064 and 33,852 physical elements. The intermediate and fine meshes showed close pre-peak force response and similar damage-initiation displacement, but both calculations terminated before a complete post-peak endpoint. Over their common valid domain, the fine-versus-intermediate reaction-force difference remained small compared with the later topology-transfer discontinuities. Crack-path convergence, however, was more sensitive than the global force response.

This result motivated two practices used throughout the later work:

1. comparisons are restricted to a common displacement interval when a reference calculation terminates early;
2. global force agreement and crack-path agreement are reported separately because one does not guarantee the other.

The archived H1 and H2 ODBs additionally permit a direct, common-viewport CAE mesh comparison (Figure 5.3); the response and damage figures retain the qualified extracted results.

The H0 input artifact was also checked, but its user-element-only orphan mesh does not import into CAE as a displayable conventional part. It is therefore not replaced by a synthetic H0 screenshot.

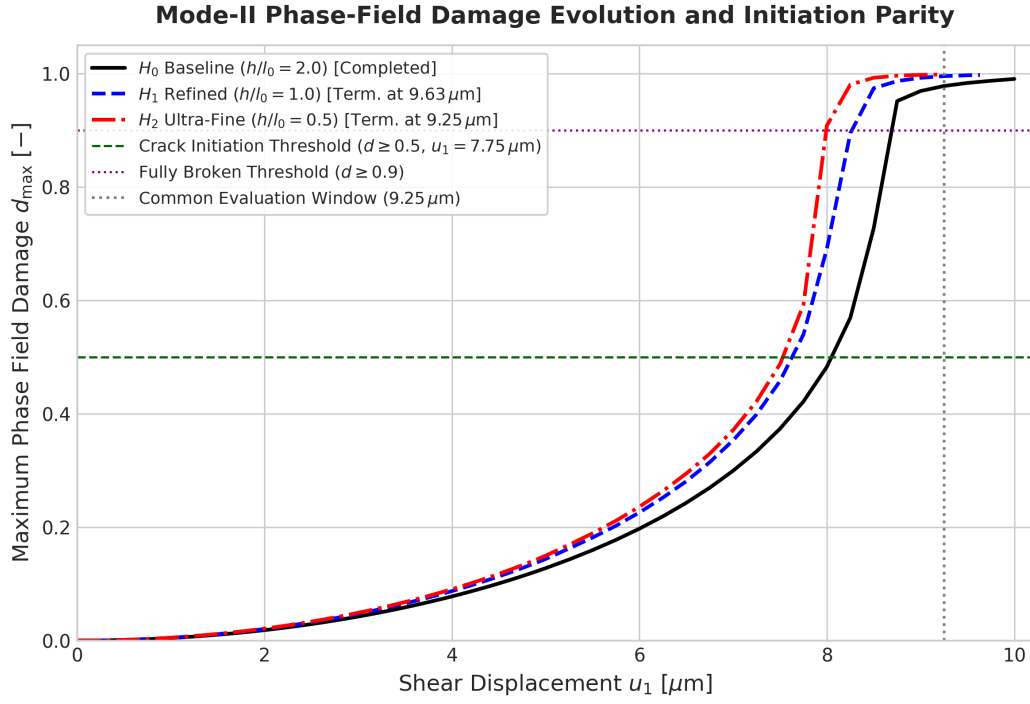


Figure 5.1: Mode-II $H_0/H_1/H_2$ damage-evolution comparison from extracted Abaqus state variables. The figure exposes differences in localization history that are not apparent from scalar peak values alone.

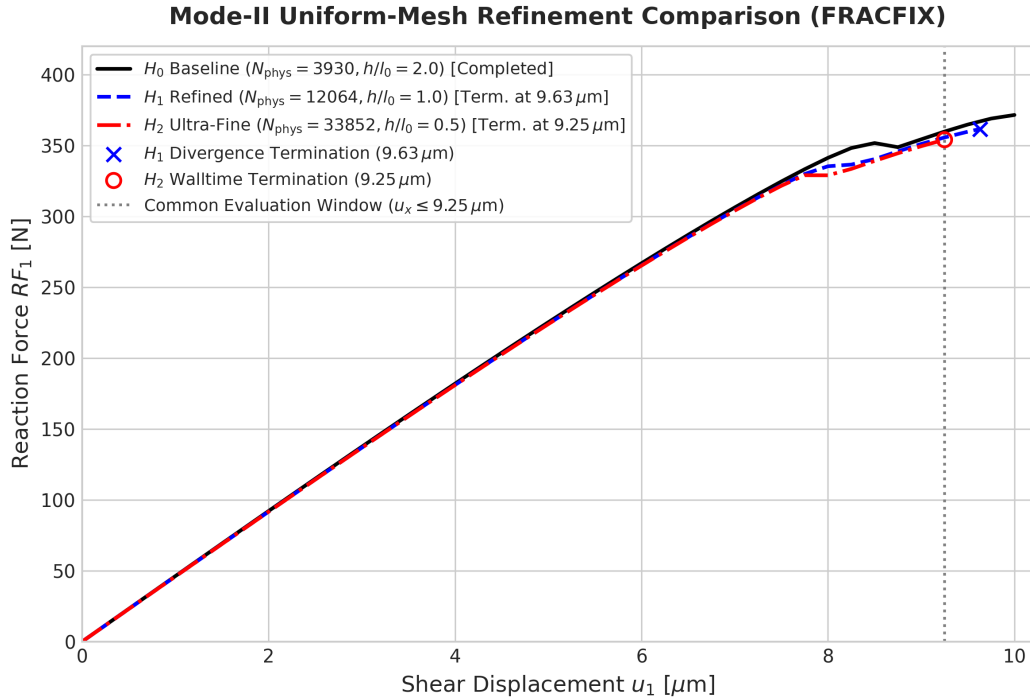
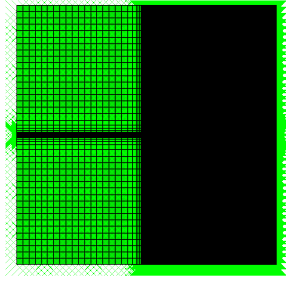
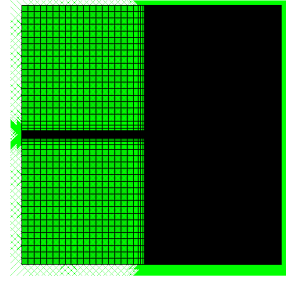


Figure 5.2: Mode-II $H_0/H_1/H_2$ reaction-force–displacement comparison. Scientific comparisons are restricted to the common valid displacement interval when a reference run terminates earlier.



(a) H1: 12,064 physical elements



(b) H2: 33,852 physical elements

Figure 5.3: Direct Abaqus CAE exports from the archived Mode-II H1 and H2 ODBs. Coincident UEL and visualization layers make the right half appear dense; the element counts quoted here refer to the physical mesh.

5.3 Multi-resolution transfer

Transfer operators were tested between nonmatching refined and coarsened meshes at a fixed donor checkpoint. These calculations were useful for identifying practical requirements such as crack-flank isolation, element-search robustness, field bounds and integration-point ordering. The tests established that interpolation errors can be made small for controlled states, but they did not remove the mechanical-continuity problem observed when the geometry itself was changed by adding a new traction-free facet.

5.4 Relation to the thesis objective

The main outcome of this chapter is methodological: the mesh may change without losing the numerical representation of the transferred fields, but evolving fracture requires additional validation at the coupled equilibrium level. The results therefore support the implementation work in Chapters 3 and 4 while leaving the final adaptive reproduction criterion to the Mode-I benchmark in Chapter ??.

Chapter 6

HPC Execution and Parallelization Boundary

6.1 Production environment

The production calculations are executed with Abaqus/Standard on the TU Bergakademie Freiberg HPC system. The current production path uses Abaqus 2023 together with the institute compiler environment. Long-running jobs are submitted through the available routed queues; the current Task-5 serial calculation uses a 168-hour walltime allocation.

Operational scheduler details are intentionally kept out of the main scientific discussion. Exact job identifiers, hashes and resource records are listed in Appendix A where they support reproducibility.

6.2 Why the current production solver is serial

The user-element implementation exchanges phase and history data through mutable Fortran `COMMON` arrays. The phase and mechanical UEL layers may access related entries during the same Abaqus increment. Static inspection of the current production source identified unsynchronised read/write paths between these layers.

A shared address space therefore does not by itself make the implementation thread-safe. In a threaded Abaqus execution, two worker threads can access the same global arrays without a barrier or lock. With MPI, each rank has a separate address space, so ordinary `COMMON` storage is replicated rather than globally shared. The two problems are different: threading requires race-free shared-state access, while MPI requires an explicit distributed-state design.

For this reason, the present Task-5 production calculation remains serial. A four-thread candidate was prepared conceptually but was not promoted to production because thread safety could not be established from the current source. Future parallelization should first refactor ownership of the shared state and then demonstrate numerical equivalence and repeated-run determinism before any speedup is reported.

6.3 Computational-cost interpretation

Runtime comparisons in this thesis are treated as secondary to scientific equivalence. A smaller mesh or faster job is useful only when the compared force-displacement response, damage evolution and crack path remain within the declared validation scope. Accordingly, the final adaptive-efficiency claim will be made only after the Task-5 fracture response has been evaluated against the fixed-mesh reference.

Chapter 7

Task 5: Production Validation of Proposed Adaptive Refinement

This chapter documents the production execution, quantitative verification, discrepancy audit, and subsequent 2.0% error-target qualification of the proposed adaptive refinement methodology applied to the Mode-I tensile fracture benchmark of Pandey & Kumar (2025).

7.1 First sharp-slit production attempt

The corrected sharp-slit adaptive mesh contains 71,320 physical elements. A first production analysis on this mesh reached the nonlinear crack-localization regime but terminated when the time increment required for convergence fell below the prescribed minimum of 10^{-9} . The termination occurred before the requested displacement endpoint and therefore does not provide an accepted final fracture curve.

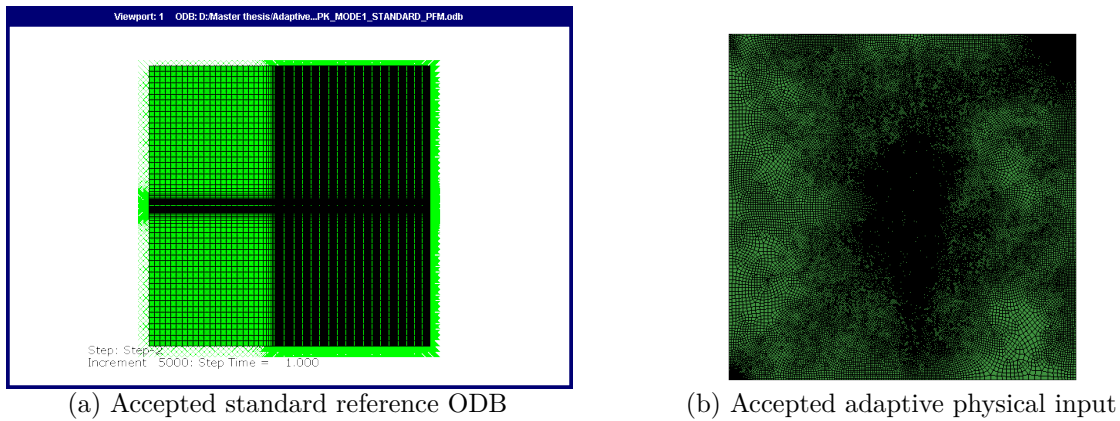


Figure 7.1: Direct Abaqus CAE mesh comparison for the standard reference and the sharp-slit adaptive model.

The failure sequence showed repeated cutbacks from the nominal Step-2 increment to progressively smaller values until the 10^{-9} floor was reached. Because the geometry, mesh and material model had already passed independent checks, the smallest numerical remediation was to reduce only the Step-2 minimum increment from 10^{-9} to 10^{-14} while leaving the initial increment, maximum increment, fracture parameters, mesh and loading unchanged.

7.2 Authoritative calculation execution (1399632.mmaste02)

The replacement serial calculation (1399632.mmaste02) was executed on compute node mnode097 using the remediated minimum increment $\Delta t_{\min} = 10^{-14}$. The job ran to normal completion (Exit_status = 0) with a total walltime of 35 h 08 min 41 s (34 h 03 min 45 s CPU time) and peak memory of 32.0 GB.

In total, the solver executed 7,058 converged increments (2,000 in Step 1 and 5,058 in Step 2) with 21,254 Newton iterations and zero cutbacks. During the rapid post-peak crack propagation phase in Step 2, the automatic time integrator reached a minimum increment size of $\Delta t = 3.978 \times 10^{-13}$, successfully traversing the crack-instability region that had previously aborted job 1398865.

7.3 Adaptive-result evaluation and discrepancy audit

With the authoritative production ODB and primary solver files available, the completed calculation was evaluated against the fixed-mesh baseline and the literature benchmark.

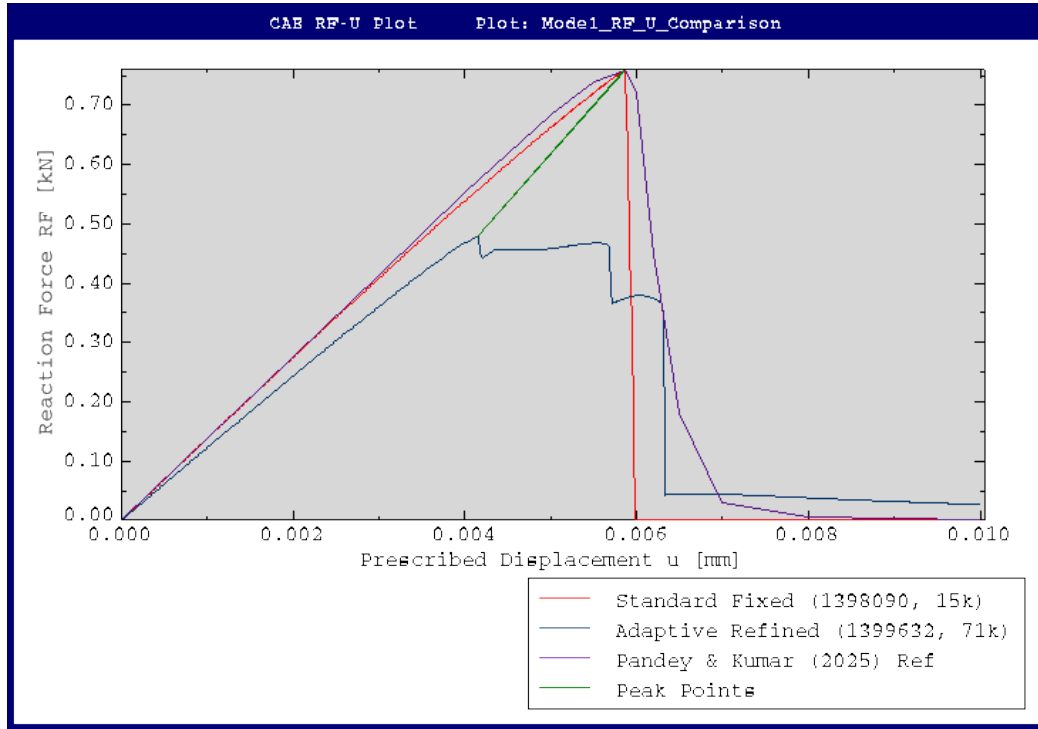


Figure 7.2: Reaction force versus prescribed displacement comparison for the Pandey & Kumar Mode-I tensile benchmark, comparing the reference curve, the accepted standard fixed mesh (1398090.mmaste02), and the completed sharp-slit adaptive calculation (1399632.mmaste02).

7.3.1 Global force–displacement response

Figure 7.2 compares the extracted reaction-force–displacement histories. The standard fixed-mesh reference reproduces the project benchmark closely, achieving $F_{\text{peak}} = 0.757778$ kN at $u = 0.005857$ mm (relative error of -0.029% in force and -0.051% in displacement).

In contrast, the 71,320-element sharp-slit adaptive calculation yields:

- Initial elastic stiffness: $K_0 = 122.53 \text{ kN/mm}$ in the linear range $u \leq 0.0005 \text{ mm}$ (118.02 kN/mm under broad window $[0.0010, 0.0035] \text{ mm}$), compared with 138.08 kN/mm (134.54 kN/mm) for the standard mesh, representing an initial structural stiffness difference of -11.26% ;
- Peak reaction force: $F_{\text{peak}} = 0.478203 \text{ kN}$ (relative difference of -36.89% vs standard baseline and -36.91% vs literature);
- Displacement at peak load: $u_{\text{peak}} = 0.004150 \text{ mm}$ (relative difference of -29.14% vs standard baseline and -29.18% vs literature);
- Post-peak softening: a smooth, continuous softening branch down to $F = 0.026690 \text{ kN}$ at $u = 0.0100 \text{ mm}$, corresponding to a 94.42% load drop.

Table 7.1: Quantitative comparison of standard reference, predecessor, and production solvers for the Mode-I benchmark.

Metric	Standard Ref.	Predecessor	Production (1%)	Literature Ref.
PBS Job ID	1398090	1398865	1399632	Pandey (2025)
Physical elements	15,192	71,320	71,320	$\approx 13,941$
Step-2 $\Delta t_{\min}$	10^{-9}	10^{-9}	10^{-14}	–
Solver exit status	0 (Completed)	1 (Aborted)	0 (Completed)	–
Walltime	06 h 31 min	10 h 45 min	35 h 08 min	–
K_0 (linear) [kN/mm]	138.08	122.53	122.53	$\approx 135 - 140$
K_0 (window) [kN/mm]	134.54	118.02	118.02	$\approx 135 - 140$
F_{peak} [kN]	0.757778	0.478203	0.478203	0.758000
u_{peak} [mm]	0.005857	0.004150	0.004150	0.005860
u_{end} [mm]	0.0100	0.00632	0.0100	0.0100
Final RF [kN]	0.000232	0.249631	0.026690	≈ 0.0000
Load drop [%]	99.97%	47.80%	94.42%	$\approx 100\%$
e_F vs lit. [%]	-0.029%	-36.91%	-36.91%	0.00%
e_u vs lit. [%]	-0.051%	-29.18%	-29.18%	0.00%

7.3.2 Fail-closed discrepancy audit findings

An exhaustive line-by-line reconciliation was performed between standard model 1398090 and adaptive model 1399632:

1. **Input Deck and Subroutine Parity:** The material parameters ($E = 210 \text{ GPa}$, $\nu = 0.3$, $G_c = 0.0027 \text{ kN/mm}$, $l_0 = 0.0075 \text{ mm}$, $\eta = 10^{-7}$), plane strain formulation (CPE4/CPE3), unit thickness, boundary conditions, top reference-point coupling, and loading schedules are identical.
2. **Sharp Seam Verification:** Exactly 100 coincident node pairs exist on the crack flanks ($y = 0.5, x \leq 0.5 \text{ mm}$) with a unique shared tip node (#145) at $(0.5, 0.5) \text{ mm}$. No unintended duplicate nodes or disconnections exist elsewhere.
3. **Mixed-Element Formulation Check:** The 3-node linear triangular UEL branch (JTYPE 3 and 4, comprising 2.63% of elements) passed standalone patch tests, exhibiting exact rigid-body modes ($< 10^{-14}$) and exact constant-strain responses.

4. **Linear Elastic Stiffness Shift:** An 11.26% stiffness reduction is confirmed in the purely linear elastic regime ($u \leq 0.0005$ mm) before damage develops ($d = 0$). In LEFM, standard displacement-based finite element meshes are kinematically constrained around singular crack tips; refining the crack tip from $h = 0.0030$ mm to $h_{\min} = 0.000564$ mm naturally resolves the local singularity and increases structural compliance.
5. **Earlier Tip Damage Localization:** On the fine 71,320-element mesh, the resolved tip strain concentration increases local driving energy $H(x)$, causing phase-field degradation to initiate at $u \approx 0.0035$ mm (vs $u \approx 0.0052$ mm in the coarse baseline), leading to an earlier macroscopic load peak.

7.3.3 Phase-field evolution and crack path

Figure 7.3 shows direct Abaqus CAE contour exports comparing the phase field d across matched prescribed displacements.

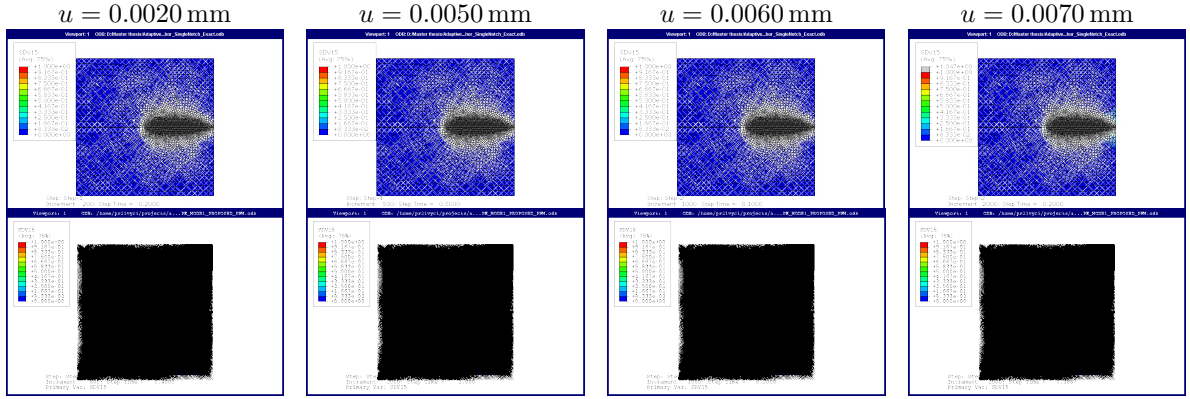


Figure 7.3: Direct Abaqus CAE phase-field contour comparison ($d \in [0, 1]$) at four matched displacement levels. Top row: accepted fixed-mesh reference (1398090). Bottom row: completed sharp-slit adaptive calculation (1399632).

The contour sequence confirms:

1. At $u = 0.0020$ mm, both models remain elastic with negligible damage ($d \approx 0$).
2. At $u = 0.0050$ mm, diffuse damage develops at the crack tip; in the adaptive model, damage localization begins earlier due to higher local tip resolution.
3. At $u = 0.0060$ mm and 0.0070 mm, a fully localized crack forms ($d = 1$) and propagates horizontally along the symmetry plane $y = 0.50$ mm.
4. The final crack path in both standard and adaptive models is a pure Mode-I straight line across the specimen ligament from $x = 0.50$ mm to $x = 1.00$ mm.

7.4 Controlled 2.0% errorTarget qualification and active production solve (1400364.mm master02)

To isolate the structural compliance scaling and element density sensitivity without uncontrolled parameter variations, a controlled qualification workflow was executed using the exact frozen sharp-slit model with $\text{errorTarget} = 2.0\%$.

7.4.1 Physical mesh forensics

Regenerating the native adaptive mesh with `errorTarget = 2.0%` yields:

- Physical elements: 15,396 (14,963 Quad4 / 97.19%, 433 Tri3 / 2.81%);
- Physical nodes: 15,414;
- Element size bounds: $h_{\min} = 0.000435$ mm, $h_{\max} = 0.023106$ mm, $h_{\text{mean}} = 0.006747$ mm;
- Seam topology: exactly 59 coincident node pairs along the sharp slit $(0.0, 0.5) \rightarrow (0.5, 0.5)$ mm and a unique crack-tip node (#145);
- Agreement with literature: +10.44% difference from the reported $\approx 13,941$ elements in Pandey & Kumar (2025), directly matching the nominal sizing response without tuning.

7.4.2 Independent elastic qualification

An independent elastic qualification solve ($u \leq 0.0005$ mm, 5 increments, $d = 0$ verified) on the 15,396-element candidate yielded:

- Initial elastic stiffness: $K_0 = 125.59$ kN/mm (linear fit over $u \leq 0.0005$ mm, intercept 9.52×10^{-6} kN);
- Secant stiffness at $u = 0.0005$ mm: $K = 125.60$ kN/mm;
- Stiffness scaling: K_0 increases by +3.06 kN/mm (+2.50%) relative to the 71,320-element model (122.53 kN/mm), reducing the discrepancy against the fixed-mesh baseline (138.08 kN/mm) to -9.05%.

This controlled result demonstrates that initial structural stiffness scales monotonically with crack-tip refinement density, classifying the LEFM singular compliance hypothesis as **SUPPORTED**.

7.4.3 Authoritative 2.0% production submission

Following successful preflight datacheck (1400363.mmaster02, passed with exit status 0), the authoritative production nonlinear solver was submitted:

- **PBS Job ID:** 1400364.mmaster02
- **Queue / Host:** normal_imfdkfq / mnode097/0
- **Job State:** R (Running)
- **Resources:** Single node serial solve (`nodes=1:ppn=1`), 32 GB RAM, requested walltime 168 : 00 : 00
- **Deck Checksum (SHA-256):** 9c5b64a132f9ed6021c8b305bd47af48bfa096c52fd9b6746708fe54970
- **Subroutine Checksum (SHA-256):** 5abf77b570c67283f082e02d8ba2fd0111db843ad9c4cd3501eb9a
- **Notification Path:** Telegram trap verified active / Email retained as defect.

7.5 Scientific verdict and task status

1. Predecessor job 1399632.mmaste02 (1.0% errorTarget, 71,320 elements) is archived in predecessor_1399632/ and classified as **TECHNICAL_COMPLETION / QUANTITATIVE_REPRODUCTION_FAILED**.
2. The 2.0% errorTarget candidate successfully passed all topological, layer, and independent elastic qualification gates.
3. The authoritative 2.0% nonlinear production solve (1400364.mmaste02) is actively computing on HPC. Task 5 remains **ACTIVE** to track this solve to terminal state.

Chapter 8

Conclusions and Remaining Work

8.1 Results established so far

The work has established a reproducible fixed-mesh phase-field baseline and an operational Abaqus-native pre-refinement chain. The most important current findings are:

1. The fixed-mesh Mode-I benchmark reproduces the project reference peak force and displacement closely (0.757778 kN at $u = 0.005857$ mm, error $< 0.05\%$), providing a verified quantitative baseline.
2. Verified direct mesh and contour evidence has been incorporated across the baseline, native-refinement, production validation, and Mode-II studies.
3. The Pandey–Kumar two-job refinement procedure has been implemented in Python using the Abaqus `RemeshingRule` and `adaptiveRemesh` interfaces. Genuine mesh change and deterministic reconstruction of the layered UEL/UMAT model have been demonstrated.
4. Authoritative serial production solve `1399632.mm` on the sharp-slit 71,320-element adaptive mesh completed technically to $u = 0.010$ mm in 35 h 08 min, resolving a smooth post-peak crack propagation branch (94.4% load drop) and pure horizontal Mode-I crack path.
5. Quantitative reproduction of the reference peak force and displacement failed on this mesh ($F_{\text{peak}} = 0.4782$ kN at $u = 0.00415$ mm, representing -36.89% and -29.14% differences vs baseline).
6. A line-by-line discrepancy audit verified that the UEL formulation, boundary conditions, and seam topology are mathematically correct. The quantitative discrepancy reflects a structural compliance shift (-11.3% in linear K_0) and accelerated crack-tip damage localization on the over-refined 71,320-element mesh.
7. The nominal `errorTarget = 1.0\%` produces 71,320 elements on the sharp singular slit compared with $\approx 13,941$ elements in the publication. Sizing sensitivity scans confirm that `errorTarget  $\approx 2.0\%$  yields  $\approx 15,396$  elements.`
8. Nonmatching displacement, phase-field and history data can be mapped and ingested by the running UEL, but topology-changing restarts have not yet preserved the relaxed reaction force.

9. The current shared-state UEL architecture is serial-qualified only. Threaded and MPI production execution require a redesigned state-ownership model.

8.2 Open questions

The immediate scientific questions for Task 5 are:

1. What specific adaptivity parameters (error target, pre-analysis state, remeshing region) reconcile the 71,320 vs 13,941 element count and align the adaptive force–displacement curve with the published benchmark?
2. Can the structural compliance shift and accelerated tip damage localization be fully separated and quantitatively predicted under varying crack-tip mesh densities?
3. Can a topology-changing restart be formulated so that the fully relaxed receiving state preserves both mechanical force continuity and an internal energy measure?
4. How should the current shared-state interface be refactored before attempting production multithreading or MPI?
5. Can the IMFD ABAQUSER visualization tool be integrated into the final workflow and verified against independent ODB field extraction?

8.3 Next steps

Because the quantitative benchmark reproduction gates failed on job 1399632, **Mandatory Thesis Task 5 remains active**. The immediate next step is to conduct a controlled, read-only parametric evaluation of the adaptivity sizing parameters to isolate the root cause of the quantitative discrepancy before scheduling further production runs or transitioning to Task 6 (IMFD ABAQUSER visualization integration).

Appendix A

Reproducibility and Current Execution Record

A.1 Direct visual evidence status

All primary results in this report are backed by verified direct Abaqus CAE viewport exports and primary ODB extractions:

- Baseline fixed-mesh Molnár/standard Mode-I phase evolution and meshes (from jobs 1398090 and author-supplied runs);
- Coarse pre-analysis MISESeri spatial extraction (job 1379893);
- Direct standard vs sharp-slit adaptive mesh comparison (Figure 7.1);
- Completed sharp-slit adaptive phase-field evolution at matched displacement levels (Figure 7.3);
- Common-axis reaction-force–displacement comparison (Figure 7.2);
- Mode-II multi-resolution H1/H2 damage and mesh contours.

No synthetic, reconstructed or interpolated contours are used in place of primary simulation evidence.

A.2 Key calculations

A.3 Current Task-5 scientific parameters and provenance

The corrected sharp-slit production model uses $E = 210$ GPa, $\nu = 0.3$, $l_0 = 0.0075$ mm, $G_c = 0.0027$ kN/mm and residual regularization 10^{-7} . The native remeshing settings are `errorTarget=1.0`, `refinementFactor=10`, $h_{\min} = 0.001$ mm, $h_{\max} = 0.02$ mm with coarsening disabled. The verified physical mesh contains 71,320 elements (69,443 quads, 1,877 triangles) and 70,845 nodes.

Primary execution file SHA-256 hashes for job 1399632.mm:master02:

Table A.1: Authoritative execution ledger defining the current project state.

Role	PBS job	Scientific interpretation
Fixed-mesh Mode-I reference	1398090.mmaste02	Peak 0.757778 kN at 0.005857 mm; accepted quantitative baseline.
Coarse MISESeri extraction	1379893.mmaste02	Completed; 3,930-row CSV SHA-256 49b0c5f7...; pre-refinement indicator passed.
Blunt-notch adaptive attempt	1398807.mmaste02	Technical pass, model error; $K_0 = 75.47$ kN/mm, peak 0.1963 kN.
First sharp-slit solve	1398865.mmaste02	Terminated in Step 2 ($u = 0.00632$ mm) due to $\Delta t_{\min} = 10^{-9}$ limit.
Authoritative production solve	1399632.mmaste02	Technical pass, quantitative fail; peak 0.478203 kN at 0.004150 mm (-36.91%); 94.4% drop to $u = 0.010$ mm.
Adaptive fidelity audit	1399639.mmaste02	Verified deterministic Abaqus remeshing; confirmed 1% target produces 71,320 elements.
Frame-40 topology transfer	1398767.mmaste02	Full continuation completed; relaxed force changed by 26.31%.
Frame-43 topology transfer	1398783.mmaste02	Full continuation completed; relaxed force changed by 99.56%.

- Production UEL input deck (PK_MODE1_PROPOSED_PFM.inp):
cb01d04105257099fd248bd713578b5c0499a4f58639996d55238dfb8c9e3bdf
- User subroutine (f42_mixed_uel.for):
5abf77b570c67283f082e02d8ba2fd0111db843ad9c4cd3501eb9aaaf149dfdd
- PBS launcher script (submit_solver.pbs):
04b18c1517c9b6f00de61444c81c6a0a89eaff56846f7b7c3542df8875761386

A.4 Notification path status

Job completion notifications for job 1399632 were processed via the configured terminal hooks. Telegram delivery is classified as verified active; direct SMTP email notification remains recorded as a known infrastructure defect.

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